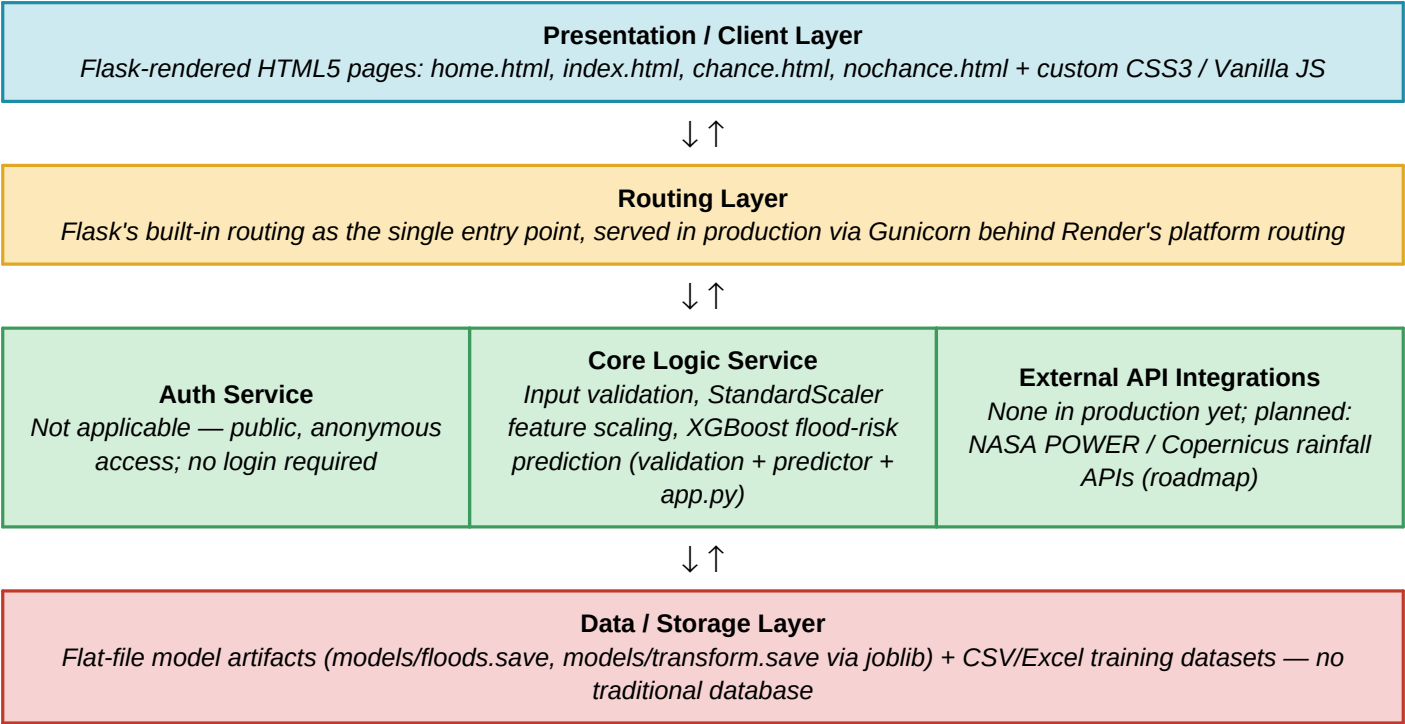


Solution Architecture

Date	15 July 2026
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

High-level view of Rising Waters' architecture: Presentation Layer, API/Routing Layer, Core Logic Layer, and Data/Storage Layer.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the landing page, prediction form, and result pages; runs client-side validation before submission.	HTML5, CSS3, Vanilla JavaScript, Jinja2 templates
Routing Layer	Flask's routing acts as the single entry point for all requests; Gunicorn serves production traffic behind Render's platform routing.	Flask, Gunicorn, Render
Core Logic Service	Validates submitted rainfall values, scales them with the saved StandardScaler, and runs the XGBoost model to produce a risk probability.	Python, scikit-learn, XGBoost, joblib, Pandas
Data / Storage	Stores the trained model and scaler as flat joblib files, plus the CSV/Excel datasets used for training; no relational or NoSQL database is used.	joblib (.save files), CSV, XLSX